

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I N.Sai Charan(192525247) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

N.Sai Charan

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS) network** to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise

resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic period

Question 1: WLAN and WWAN Optimisation for a Smart Office Building

1. Problem Identification

Analysis of the scenario reveals a combination of RF-layer, access-layer, and security-layer faults across both the indoor WLAN and the field WWAN:

- Uneven access point (AP) loading — some APs are overloaded while neighbouring APs are idle, indicating poor AP placement and absence of load balancing/band steering.
- Floor-to-floor disconnections — likely caused by weak or missing fast-roaming support (802.11r/k/v), so client devices cannot hand off seamlessly between APs as users move.
- Weak signal in conference rooms — RF attenuation from concrete/glass partitions in the six-storey building, combined with insufficient AP density in high-usage rooms.
- Video conferencing freezing — no QoS prioritisation (WMM/DSCP marking) for real-time traffic, so bulk downloads and video/voice contend for the same airtime.
- Peak-hour slowdowns — channel congestion from co-channel/adjacent-channel interference (limited non-overlapping 2.4 GHz channels: 1, 6, 11) combined with high client density (1,200+ devices).
- Field engineer WWAN drop-outs — coverage gaps in rural/highway areas are inherent to carrier cellular footprint, compounded by absence of multi-carrier failover or offline data caching.
- Security exposure — some devices are still authenticating with outdated protocols (WEP/WPA-PSK), which are vulnerable to dictionary and key-recovery attacks.

2. Debugging Process

1. Conduct a full RF site survey (predictive and on-site) per floor to generate heat maps and identify dead zones, co-channel interference, and AP overlap.
2. Capture wireless traffic (e.g., with Wireshark/AirMagnet) at problem locations to inspect retransmission rates, deauthentication frames, and channel utilisation.
3. Review the WLAN controller logs for per-AP client counts, RSSI values, and roaming/hand-off events to confirm load imbalance and roaming failures.
4. Audit the channel and transmit-power plan across all APs to check for overlapping channel reuse and unnecessary maximum-power settings that widen interference.
5. Audit SSID and authentication configuration on every AP to flag devices/APs still permitting WEP or WPA-PSK connections.
6. Collect WWAN signal logs and application-level upload logs from field devices to correlate report-upload failures with specific geographic coverage gaps.

3. Optimisation Strategies

- Redesign AP placement using survey data — apply cell-splitting (more APs at lower transmit power) so load is spread evenly and coverage in conference rooms is reinforced.

- Enable controller-based load balancing and band steering so capable clients are steered to the less congested 5 GHz band automatically.
- Enable 802.11r (fast BSS transition), 802.11k, and 802.11v so devices roam between floors without dropping active sessions.
- Enable Radio Resource Management (RRM) for dynamic channel and power assignment, eliminating manual co-channel interference.
- Configure QoS (WMM classes and DSCP tagging) to prioritise VoIP and video conferencing traffic over bulk file transfers.
- Migrate all devices to WPA3-Enterprise/802.1X with a RADIUS server and disable legacy WEP/WPA-PSK associations.
- For field staff, deploy multi-carrier/bonded WWAN modems with automatic failover, and use store-and-forward mobile applications that cache maintenance reports locally and sync once connectivity is restored.
- Evaluate private 5G/CBRS or additional carrier-grade small cells for consistently weak highway/rural corridors used by field engineers.

4. Analysis and Justification

The uneven AP load and floor hand-off failures both stem from a static, unmanaged RF design; RRM together with fast-roaming standards addresses the root cause rather than symptomatically adding more APs. Prioritising real-time traffic through QoS directly targets the video-freezing complaint, since the underlying capacity is often sufficient once contention is managed. Replacing legacy authentication closes the security gap identified in the audit without requiring new hardware. For WWAN, because carrier coverage is outside the company's direct control, the optimisation focus shifts to resilience (multi-carrier failover) and graceful degradation (offline caching) rather than eliminating gaps outright — this is the most cost-effective and realistic mitigation for rural/highway connectivity.

Question 2: ATM Network Performance Debugging and Optimisation

1. Problem Identification

- High utilisation on several ATM switches during peak hours, indicating the physical/link layer is nearing saturation at specific nodes.
- Congested Virtual Circuits (VCs) — bandwidth is not being efficiently allocated across VCs, so some circuits are starved while others are lightly used.
- Applications are assigned inappropriate ATM service categories (e.g., real-time voice/video mapped onto UBR or nrt-VBR instead of CBR/rt-VBR), directly causing jitter, delay and packet loss for VoIP and video.
- Delays in online banking transactions suggest Connection Admission Control (CAC) is permitting more traffic onto a VC than its negotiated Peak/Sustainable Cell Rate (PCR/SCR) can support.

2. Debugging Process

7. Pull per-VC performance counters from ATM switches — Cell Loss Ratio (CLR), Cell Delay Variation (CDV), and Cell Transfer Delay (CTD) — to pinpoint which VCs are out of specification.
8. Cross-check the service category configured on each VC against the actual traffic profile it carries (voice, video, or bulk data).
9. Trace congestion points using ATM network management tools to identify whether the bottleneck is at the switch fabric, trunk link, or specific VC buffer.
10. Verify CAC parameters against real subscribed bandwidth to detect over-subscription of high-priority VCs.
11. Correlate customer-reported call/video-quality issues with the specific VC and switch handling that traffic at the time of the complaint.

3. Optimisation Strategies

- Re-map traffic to the correct ATM service category: CBR for voice, real-time VBR (rt-VBR) for interactive video, non-real-time VBR (nrt-VBR)/ABR for transactional data, and UBR only for best-effort background traffic.
- Apply traffic shaping and policing at the User-Network Interface using the Generic Cell Rate Algorithm (leaky-bucket) so each VC conforms to its negotiated PCR/SCR/MBS.
- Rebalance VCs across underutilised switches and trunks through traffic engineering, and add trunk capacity or link aggregation where sustained utilisation is consistently high.
- Tighten CAC so that new VC requests are only accepted when sufficient bandwidth remains to meet the QoS guarantees of existing VCs.
- Deploy continuous SNMP-based QoS monitoring (CLR/CDV dashboards) so congestion is caught proactively rather than after customer complaints.

4. Analysis and Justification

ATM's fixed 53-byte cell size exists precisely to bound delay and jitter for real-time traffic — but this benefit is only realised when each application uses the service category matched to its traffic characteristics. Misclassification (e.g., voice on UBR) discards ATM's core QoS advantage, which explains the reported jitter and packet loss despite the network's deterministic design. Correcting service-category mapping and enforcing CAC/traffic shaping restores the predictable QoS the ATM architecture is designed to provide, while traffic engineering across underused switches resolves the uneven-utilisation finding without requiring new core hardware.

Question 3: MPLS Network Fault Debugging and Optimisation

1. Problem Identification

- Some Label Switched Paths (LSPs) are heavily congested while others remain underutilised — a clear sign that traffic engineering (TE) is either absent or not constraint-based.
- Incorrect label mappings at several routers cause inefficient routing paths and added latency, consistent with LDP/RSVP-TE misconfiguration.
- Poor-quality voice calls and video interruptions indicate no Class of Service (CoS)/EXP-bit based prioritisation is applied across the MPLS backbone.
- Delayed order processing under load suggests no Fast Reroute (FRR) or rapid convergence mechanism exists for handling path failures or reroutes.

2. Debugging Process

12. Inspect the MPLS label forwarding tables (LFIB) and LDP/RSVP-TE session status on each Label Switch Router to confirm session health and label consistency.
13. Run MPLS LSP ping and traceroute hop-by-hop to detect where label swaps deviate from the expected path or where a label mapping is stale/incorrect.
14. Analyse backbone link-utilisation statistics to map which LSPs are congested and which are idle.
15. Review the MPLS-TE tunnel configuration and Constrained Shortest Path First (CSPF) parameters to see whether bandwidth constraints are actually being considered during path selection.
16. Check EXP-bit/CoS marking on VoIP, video, and ERP traffic entering the MPLS core to confirm whether classification is applied at the ingress PE router.

3. Optimisation Strategies

- Enable MPLS-TE with CSPF so LSPs are computed based on real available bandwidth rather than shortest IGP path alone, redistributing load away from congested links.
- Correct and standardise label bindings across all routers, ensuring consistent LDP/RSVP-TE timers and parameters network-wide.
- Implement CoS-based traffic engineering: mark VoIP with the highest priority EXP value (strict priority queue), video/ERP with guaranteed-bandwidth classes, and bulk data as best-effort.
- Enable MPLS Fast Reroute so traffic switches to a pre-computed backup LSP within milliseconds of a link/node failure, avoiding user-visible interruptions.
- Where supported, load-share traffic across multiple parallel LSPs (unequal-cost multipath) instead of concentrating all traffic on a single primary path.
- Introduce an SDN/NMS-based traffic-engineering controller to continuously monitor and rebalance LSP assignments as demand shifts throughout the day.

4. Analysis and Justification

MPLS's advantage over pure IP routing is that a path can be explicitly engineered rather than determined solely by shortest-path IGP metrics; the uneven LSP utilisation observed here shows that advantage is not being used. Introducing CSPF-based TE directly fixes the imbalance, while correcting label bindings removes the latency caused by inefficient hop sequences. Voice and video quality issues are addressed at the classification layer (EXP/CoS) rather than by adding bandwidth, since the core issue is prioritisation, not raw capacity. Fast Reroute closes the remaining gap around convergence delay during link failures, which is critical for an e-commerce backbone carrying time-sensitive order-processing traffic.

Question 4: Error Detection and Correction in an Automated Warehouse Network

1. Problem Identification

- Shipment records arriving with incorrect data and failed barcode scans point to undetected bit errors reaching the application layer.
- Electrical interference (from motors, robots, conveyor systems) and signal attenuation are introducing burst errors, which weak error-detection schemes (e.g., simple parity) cannot reliably catch.
- Some corrupted frames are retransmitted successfully, but others are not — implying the retransmission (ARQ) mechanism is either mistimed or not triggered for all detected errors.
- Rising transmission errors during peak hours also point to congestion-related collisions on a shared wireless channel used by many scanners and robots simultaneously.

2. Debugging Process

17. Capture frame-level statistics (CRC/checksum failure counts, frame error rate) at different times of day and warehouse zones to correlate errors with peak-hour congestion and specific EMI sources.
18. Perform an RF/EMI scan near motors, conveyors, and robot charging stations to identify physical interference hotspots affecting wireless scanners.
19. Review the current error-detection scheme in use (parity bit, simple checksum, or CRC) and evaluate whether its polynomial/algorithm can reliably catch burst errors of the length observed.
20. Inspect ARQ logs (Stop-and-Wait/Go-Back-N/Selective-Repeat) to check retransmission timeout values and whether timeouts are firing correctly under congestion.
21. Test whether corrupted frames are passing a weak checksum undetected (false negative), by deliberately injecting known bit errors in a controlled test and observing whether the checksum flags them.

3. Optimisation Strategies

- Upgrade error detection from parity/simple checksum to CRC-32, which detects far more burst-error patterns and is standard for industrial data links.
- Introduce Forward Error Correction (e.g., Hamming code for single-bit correction, or Reed-Solomon for burst errors) so many corrupted frames are corrected in real time without needing a retransmission at all.
- Switch from Stop-and-Wait ARQ to Selective-Repeat ARQ so only the specific corrupted frame is retransmitted, improving throughput under the observed congestion.
- Address the physical layer directly: shield cabling, relocate or add access points away from motor/robot EMI sources, and use industrial-grade, EMI-hardened equipment.
- Use frequency-hopping spread spectrum (FHSS) for wireless scanners/RFID readers to reduce the impact of narrowband interference sources.

- Segment the warehouse network into smaller VLANs/collision domains with additional access points to reduce contention during peak scanning periods.
- Add an end-to-end application-layer checksum for inventory/shipment records as a defence-in-depth measure, so any error that slips past the link layer is still caught before the database is updated.

4. Analysis and Justification

Simple parity can only reliably detect single-bit errors, but the described environment (motors, conveyors, EMI) is a classic source of multi-bit burst errors, which explains why some corrupted frames go undetected. Moving to CRC-32 closes this detection gap, and layering FEC on top reduces how often retransmission is even needed — directly improving throughput during peak congestion. Selective-Repeat ARQ ensures that when retransmission is still necessary, it is efficient rather than re-sending whole windows of otherwise-good frames. Because interference is physical in origin, purely protocol-level fixes (better CRC/ARQ) are necessary but not sufficient; the RF/EMI hardening steps address the root cause rather than only its symptoms.

Question 5: Flow Control Debugging and Optimisation for an Online Examination Platform

1. Problem Identification

- Receiver buffer overflow and application crashes on student devices show that the sender is transmitting faster than slower clients (broadband/mobile) can consume data — a classic flow-control mismatch.
- Incomplete downloads and repeated retransmissions suggest the sliding-window/acknowledgement mechanism is not adapting to each client's actual capacity (a fixed or oversized window is likely in use).
- Duplicate packets point to a retransmission-timeout mechanism that is firing before the original packet's acknowledgement has had time to arrive, especially over higher-latency mobile links.
- The large asymmetry between the server's 10 Gbps link and students' much slower connections is the structural root cause: a single flow-control policy cannot serve both ends of that gap efficiently.

2. Debugging Process

22. Capture traffic during a live exam window and measure retransmission counts, duplicate ACKs, and RTT variance between mobile and broadband clients.
23. Correlate receiver buffer-overflow/crash reports with the throughput bursts recorded on the server side at the same timestamps.
24. Review the current flow-control configuration (e.g., TCP window size, whether window scaling is enabled) to confirm if the advertised receive window is actually being honoured.
25. Reproduce the fault in a controlled network-emulation test with simulated slow/high-latency clients to confirm buffer overflow occurs under those specific conditions.
26. Check whether acknowledgements are cumulative-only or support Selective Acknowledgment (SACK), since cumulative-only ACKs under loss cause unnecessary full-window retransmissions.

3. Optimisation Strategies

- Enable TCP window scaling and adaptive sliding-window flow control so the sender's transmission rate is matched to each individual receiver's advertised window rather than a single fixed rate.
- Enable Selective Acknowledgment (SACK) so only genuinely lost segments are retransmitted, instead of the entire outstanding window.
- Introduce application-level backpressure: the server should wait for a client-ready/ACK signal for the previous data chunk before releasing the next, especially for large media (images/videos) in the question paper.
- Deploy CDN edge caching / regional distribution nodes for static exam content (images, videos) so the origin server is not the single choke point for 15,000+ simultaneous students.
- Serve adaptive-quality content — lower-resolution versions of images/videos to clients detected on slow mobile connections — to reduce the data volume that must be flow-controlled in the first place.

- Tune congestion-control algorithm (e.g., adopt BBR or CUBIC with appropriate parameters) to handle the wide RTT variance between mobile and broadband clients more gracefully than default settings.
- Scale the serving tier horizontally (multiple front-end servers behind a load balancer/queue) so the fixed 10 Gbps origin does not have to fan out to every student directly.

4. Analysis and Justification

Flow control exists to prevent exactly the symptom described — a fast sender overwhelming a slow receiver's buffer — so the fix must target rate-matching, not just bandwidth. Enabling window scaling and honouring each client's real advertised window is the direct remedy for buffer overflow. SACK reduces the retransmission volume created by a lossy or high-latency link without requiring more bandwidth. Because the root asymmetry is structural (10 Gbps origin vs. heterogeneous last-mile links), protocol tuning alone is not enough; distributing content via CDN edge nodes and serving adaptive quality reduces the absolute data volume each client must flow-control, which is the most durable fix for a 15,000-student concurrent load.